Exercise #1

$$A = \begin{pmatrix} -b_1 & b_2 \\ b_3 & -b_1 \end{pmatrix}$$

$$A - \lambda I = 0$$

$$\begin{pmatrix} -b_1 & b_2 \\ b_3 & -b_1 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} =$$

$$\begin{pmatrix} -b_1 - \lambda & b_2 \\ b_3 & -b_1 - \lambda \end{pmatrix}$$

$$\det(A - \lambda I) = 0$$

$$(-b_1 - \lambda)^2 - (b_2 b_3) = 0$$

$$(b_1 + \lambda)^2 = b_2 b_3$$

$$\lambda = \pm\sqrt{b_2 b_3} - b_1$$

$$\lambda_1 = \sqrt{b_2 b_3} - b_1 \qquad \lambda_2 = -\sqrt{b_2 b_3} - b_1$$

<u>Case 1:</u> both λ's are neg.

Since $b_1, b_2, b_3 > 0 \Rightarrow \lambda_2$ always neg.

to get λ_1 neg.

$$\hookrightarrow \sqrt{b_2 b_3} - b_1 < 0$$

$$b_2 b_3 < b_1^2$$

<u>Case 2:</u> one λ pos., other λ neg.

λ_2 always neg., so $\lambda_1 > 0$

$$\sqrt{b_2 b_3} - b_1 > 0$$

$$b_2 b_3 > b_1^2$$